A Triangle with a Three-Edge Tail: An Odd-Walk Domination Proof

Abstract

Let R_3 be a triangle with a path of length three attached at one triangle vertex. We prove that every graphon W of edge density $p \geq 1/2$ satisfies

$$t(R_3, W) \geq p^4(2p-1) = \Phi_{R_3}(p).$$

The new input is the special path-density inequality

$$t(P_5, W)^3 \geq t(P_3, W)^5,$$

where P_j denotes the path with j edges. This is the $(3, 5)$ case of the Erdős–Simonovits odd-walk theorem proved by Blekherman and Raymond. We give a complete transfer from finite hosts to graphons on arbitrary probability spaces, then combine the path ratio with the pointwise rooted triangle bound. The result classifies the formerly open Atlas graph 102.

1 Graphon notation and the graph

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be a symmetric measurable graphon. Write

$$p = t(K_2, W), \qquad d = T_W\mathbf{1}, \qquad A = T_W d = T_W^2\mathbf{1}, \qquad B = T_W A = T_W^3\mathbf{1}.$$

Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Let R_3 have a triangle rooted at o and three additional vertices u, v, w, with the only additional edges ou, uv, vw. Integrating the tail from its free end toward the root gives

$$t(R_3, W) = \int_\Omega \tau(x)B(x)\,\mathrm{d}\mu(x). \tag{1}$$

It has six vertices and six edges. Every vertex added along the tail has $x-1$ colour choices after its parent is coloured, so

$$\chi_{R_3}(x) = x(x-1)^4(x-2), \qquad \chi(R_3) = 3.$$

Consequently

$$\Phi_{R_3}(p) = p^4(2p-1). \tag{2}$$

2 The two path inequalities on arbitrary graphons

Let P_j denote the path with j edges and $j+1$ vertices. We need two standard walk inequalities in a form that applies to the probability-space generality used throughout the catalogue.

Lemma 2.1 (Blakley–Roy for P_3)**.** *Every graphon of edge density p satisfies*

$$t(P_3, W) \geq p^3.$$

Proof. Since $d(x) = 0$ implies $W(x, y) = 0$ for almost every y, define quotients by $d(x)$ to be zero on that null support. Almost everywhere on the support of W,

$$W = (W dd')^{1/3}(W/d)^{1/3}(W/d')^{1/3},$$

where $d = d(x)$ and $d' = d(y)$. Hölder gives

$$p^3 \leq \left(\int_{\Omega^2} W(x,y)d(x)d(y)\,\mathrm{d}\mu^2\right)\left(\int_{\Omega^2} \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu^2\right)\left(\int_{\Omega^2} \frac{W(x,y)}{d(y)}\,\mathrm{d}\mu^2\right).$$

Each quotient integral is at most one. The first factor is exactly $t(P_3, W)$. □

Lemma 2.2 (The $(3, 5)$ odd-walk inequality)**.** *Every graphon W on every probability space satisfies*

$$t(P_5, W)^3 \geq t(P_3, W)^5.$$

Proof. For finite simple host graphs this is the $t = 3, k = 5$ instance of the Erdős–Simonovits odd-walk theorem; we use the proof of Blekherman and Raymond [1]. We now justify the graphon transfer without assuming that the underlying probability space is atomless or standard Borel.

Fix a finite graph F. For each N, sample independent points $X_1, \ldots, X_N$ from μ. Conditional on these points, and independently for every unordered pair $i < j$, place the edge ij in a simple graph G_N with probability $W(X_i, X_j)$. Auxiliary independent uniform random variables on $[0, 1]$ realize these Bernoulli choices.

Let $T_N(F) = t(F, G_N)$. Among the $N^{v(F)}$ maps $V(F) \to [N]$, at most $\binom{v(F)}{2}N^{v(F)-1}$ are non-injective. For an injective map, all host edges used by distinct edges of F are distinct; conditional independence and then integration of the sampled points show that its expected indicator is exactly $t(F, W)$. Hence

$$|\mathbb{E}T_N(F) - t(F, W)| \leq \frac{\binom{v(F)}{2}}{N}. \tag{3}$$

For the variance, expand $T_N(F)^2$ as a sum over two vertex maps. If the two image sets are disjoint, their indicators depend on disjoint sampled points and disjoint auxiliary edge variables, and are independent. The number of ordered pairs of maps with intersecting images is at most $v(F)^2 N^{2v(F)-1}$. Every covariance has absolute value at most one, so

$$\operatorname{Var} T_N(F) \leq \frac{v(F)^2}{N}. \tag{4}$$

Equations (3)–(4) imply $T_N(F) \to t(F, W)$ in probability.

Apply this simultaneously to $F = P_3, P_5$. Every realization of the finite simple graph G_N satisfies

$$T_N(P_5)^3 \geq T_N(P_3)^5.$$

If the claimed limiting inequality were false, continuity would give an open neighbourhood of $(t(P_3, W), t(P_5, W))$ on which the reverse strict inequality holds. Simultaneous convergence in probability places $(T_N(P_3), T_N(P_5))$ in that neighbourhood with probability tending to one, contradicting the finite inequality. The graphon statement follows. □

Combining the preceding lemmas gives the ratio in precisely the form needed below.

Corollary 2.3 (Odd-path weighted mean)**.** *If $p > 0$, put $Z = t(P_3, W)$. Then $Z > 0$ and*

$$\frac{t(P_5, W)}{Z} \geq p^2.$$

Proof. Lemma 2.1 gives $Z \geq p^3 > 0$. Lemma 2.2 gives

$$\frac{t(P_5, W)}{Z} \geq Z^{2/3} \geq p^2.$$

□

3 The positive theorem

Lemma 3.1 (Pointwise rooted triangle)**.** *For almost every x,*

$$\tau(x) \geq \max\{2A(x) - p, 0\}.$$

Proof. Apply $ab \geq a + b - 1$ to $W(x, y), W(x, z)$, multiply by $W(y, z) \geq 0$, and integrate in y, z. This gives $\tau(x) \geq 2A(x) - p$; also $\tau(x) \geq 0$. □

Theorem 3.2 (Triangle with a three-edge tail)**.** *For every graphon W of edge density $p \in [1/2, 1]$,*

$$\boxed{t(R_3, W) \geq p^4(2p - 1) = \Phi_{R_3}(p).}$$

Proof. Put

$$Z = \int_\Omega B(x)\, \mathrm{d}\mu(x).$$

The operator definitions and self-adjointness of T_W give

$$Z = \langle \mathbf{1}, T_W^3 \mathbf{1} \rangle = t(P_3, W), \tag{5}$$

$$\int_\Omega B(x)A(x)\, \mathrm{d}\mu(x) = \langle T_W^3 \mathbf{1}, T_W^2 \mathbf{1} \rangle = \langle \mathbf{1}, T_W^5 \mathbf{1} \rangle = t(P_5, W). \tag{6}$$

Since $p \geq 1/2 > 0$, Lemma 2.1 gives $Z > 0$. Define the probability measure

$$\mathrm{d}\nu(x) = \frac{B(x)}{Z}\, \mathrm{d}\mu(x).$$

By (5), (6), and Corollary 2.3,

$$\mathbb{E}_\nu A = \frac{t(P_5, W)}{t(P_3, W)} \geq p^2. \tag{7}$$

The function $f(a) = \max\{2a - p, 0\}$ is convex and nondecreasing. Using (1), Lemma 3.1, Jensen, (7), and $p^2 \geq p/2$, we obtain

$$\begin{aligned} t(R_3, W) &\geq Z\, \mathbb{E}_\nu f(A) \geq Z f(\mathbb{E}_\nu A) \geq Z f(p^2) \\ &= Z(2p^2 - p) \geq p^3\, p(2p - 1) = p^4(2p - 1). \end{aligned}$$

The last expression is the target in (2). □

At $p = 1/2$ the target is zero, and the proof remains valid without division by $2p - 1$. At $p = 1$, $W = 1$ almost everywhere and both sides are one.

4 Atlas consequence and formalization boundary

Corollary 4.1 (Atlas 102)**.** *NetworkX Atlas graph* 102*, with graph6 code* `EJe?`*, is positive. It has order and size six, chromatic number three, and*

$$\chi_{F_{102}}(x) = x(x-1)^4(x-2).$$

For every graphon of edge density $p \in [1/2, 1]$*,*

$$t(F_{102}, W) \geq p^4(2p-1) = \Phi_{F_{102}}(p).$$

Proof. Its NetworkX edge list is

$$04,\ 05,\ 12,\ 13,\ 23,\ 34.$$

The vertices $1, 2, 3$ form a triangle, and the remaining edges form the three-edge path $3-4-0-5$. Thus Atlas 102 is exactly R_3. □

The proof works with ordinary, non-injective homomorphism densities. The sampling transfer explicitly controls collisions and variance and therefore does not silently replace homomorphism counts by injective counts. It also does not require the graphon probability space to have atoms, to be atomless, or to be standard Borel.

For formalization, the genuinely new infrastructure is the special odd-walk comparison in Lemma 2.2. One may formalize either the finite Blekherman–Raymond $(3, 5)$ theorem together with the sampling transfer, or a direct analytic proof of that special case. After that lemma, the graphon proof uses only operator-coordinate peeling, the already formalized P_3 bound, the existing pointwise rooted-triangle inequality, and scalar Jensen for the positive part of an affine function.

This argument applies specifically because a single three-edge tail produces the consecutive odd path ratio $t(P_5)/t(P_3)$. It does not prove arbitrary pendant-tree closure, multiple three-edge tails, a three-edge broom, or tails attached at several triangle vertices. In particular it does not classify Atlas 97 or 100.

A A shorter route, and a smaller gap

This appendix was written while preparing the Lean formalization. It records two reductions of the argument above and one obstruction, and it changes nothing about what is proved. Throughout write

$$a_k = t(P_k, W) = \langle \mathbf{1}, T_W^k \mathbf{1} \rangle, \qquad a_0 = 1, \quad a_1 = p.$$

A.1 The convexity step is not needed

In the proof of Theorem 3.2 the function $f(a) = \max\{2a - p, 0\}$ is only ever integrated against $B \geq 0$. Since $f(a) \geq 2a - p$ pointwise, (1) and Lemma 3.1 already give

$$t(R_3, W) = \int_\Omega \tau B \, \mathrm{d}\mu \ \geq \ \int_\Omega (2A - p) B \, \mathrm{d}\mu \ = \ 2\int_\Omega AB \, \mathrm{d}\mu - p \int_\Omega B \, \mathrm{d}\mu \ = \ 2a_5 - p\, a_3, \tag{8}$$

by (5) and (6). No auxiliary probability measure ν, no Jensen, and no convexity or monotonicity of f is used.

A.2 The odd-walk theorem is stronger than the theorem needs

By (8), Theorem 3.2 reduces to the scalar statement $2a_5 - p\,a_3 \geq p^4(2p-1)$, equivalently

$$2\big(a_5 - p^5\big) \;\geq\; p\big(a_3 - p^3\big). \tag{9}$$

Lemma A.1. *Suppose $p \geq 1/2$ and*

$$a_5 \;\geq\; p^2 a_3. \tag{10}$$

Then $t(R_3, W) \geq p^4(2p-1)$.

Proof. By (8) and (10), $t(R_3, W) \geq 2a_5 - p\,a_3 \geq (2p^2 - p)a_3 = p(2p-1)a_3$. Since $2p - 1 \geq 0$ and $a_3 \geq p^3$ by Lemma 2.1, this is at least $p(2p-1)p^3 = p^4(2p-1)$. □

Inequality (10) is strictly weaker than Lemma 2.2: from $a_5^3 \geq a_3^5$ and $a_3 \geq p^3$,

$$a_5 \;\geq\; a_3^{5/3} = a_3 \cdot a_3^{2/3} \;\geq\; a_3 p^2.$$

So the whole formalization gap is (10), one inequality among three walk densities, rather than the Erdős–Simonovits odd-walk theorem together with a sampling transfer. Every other ingredient is already mechanically checked: Lemma 2.1 is `pow_three_le_pathIntegral`, Lemma 3.1 is `rootedTriangle_ge`, and (1) follows from the assignment-integral peeling ladder.

A.3 What (10) is not

Two remarks delimit the search, both negative.

First, (10) is not a statement about the spectral measure alone. Let σ be the spectral measure of $\mathbf{1}$ for T_W, so that $a_k = \int \lambda^k \,\mathrm{d}\sigma$ with σ a probability measure. The inequality $\int \lambda^5 \geq (\int \lambda)^2 \int \lambda^3$ is false for general probability measures: for $\sigma = \frac{1}{17}\delta_{-1} + \frac{16}{17}\delta_{1/2}$,

$$\int \lambda^3 \,\mathrm{d}\sigma = \tfrac{1}{17} > 0, \qquad \int \lambda^5 \,\mathrm{d}\sigma = -\tfrac{1}{34} < 0.$$

Adding an atom of small mass at $+1$ makes σ satisfy the Perron–Frobenius constraint $\operatorname{supp}\sigma \subseteq [-\lambda_{\max}, \lambda_{\max}]$ with $\sigma(\{\lambda_{\max}\}) > 0$, and still leaves $\int \lambda^5 < 0 < (\int \lambda)^2 \int \lambda^3$. Hence no argument using only the moment data of σ can succeed: nonnegativity of W has to enter somewhere other than through the spectrum. What it gives for free is only $a_{2k+1} = \langle T_W^k \mathbf{1}, T_W\, T_W^k \mathbf{1}\rangle \geq 0$.

Second, four natural routes fail, each of them on the complete bipartite graphon W_s with parts of measure s and $1-s$, for which

$$p = 2s(1-s), \quad a_2 = s(1-s), \quad a_3 = 2s^2(1-s)^2, \quad a_4 = s^2(1-s)^2, \quad a_5 = 2s^3(1-s)^3.$$

1. *Hölder in the Blakley–Roy pattern.* Factoring $W\,dd' = (WAA')^{1/3}\big(W d^3 d'^3/(AA')\big)^{1/3} W^{1/3}$ and applying Hölder with exponents $(3,3,3)$ gives $a_3^3 \leq a_5 \cdot J \cdot p$, where $J = \iint W\,(d^3/A)(d'^3/A')$. One would need $J \leq p^3$. On W_s, $J = 2s^2(1-s)^2$ while $p^3 = 8s^3(1-s)^3$, so $J \leq p^3$ forces $(1-2s)^2 \leq 0$.

2. *Hölder with exponents* $(5/3, 5/2)$. This reduces (10) to $\langle h, T_W h\rangle \leq 1$ for $h = d^{5/2}/A^{3/2}$. On every regular graphon $\langle h, T_W h\rangle = 1$ exactly, so the reduced statement is tight and no crude estimate closes it.

3. *A positive-definite correction.* With $u = A - pd = T_W(d-p)$, $\langle u, T_W u\rangle = a_5 - 2pa_4 + p^2 a_3$. Were this nonnegative, (10) would follow from $a_4 \geq p\, a_3$, itself a consequence of $a_3 \leq (a_2 a_4)^{1/2}$, $a_4 \geq a_2^2$ and $a_2 \geq p^2$. But on W_s one has $u = \mp s(1-s)(1-2s)$ on the two parts and $\langle u, T_W u\rangle = -2s^3(1-s)^3(1-2s)^2 < 0$ for $s \neq 1/2$.

4. *Tilting.* Writing $a_5 = a_2^2 p_A$ and $a_3 = p^2 p_d$, where p_A and p_d are the edge densities of W with respect to $A\,\mathrm{d}\mu/a_2$ and $d\,\mathrm{d}\mu/p$, inequality (10) would follow from $a_2 \geq p^2$ together with $p_A \geq p_d$. The latter is $a_5 p^2 \geq a_3 a_2^2$, which on W_s reads $4s(1-s) \geq 1$.

A.4 A stronger statement consistent with all numerical evidence

The following would give (10) through $a_5 \geq a_2 a_3 \geq p^2 a_3$, and would also contain Lemma 2.1 through $a_3 \geq a_1 a_2 \geq a_1^3$.

Conjecture A.2. *For every graphon W and all $m, n \geq 0$, $a_{m+n} \geq a_m a_n$.*

Equality holds throughout whenever W is regular, since then $a_k = p^k$. The statement was tested on roughly 5×10^6 pseudorandom graphons — uniform $[0,1]$ kernels, Bernoulli 0/1 kernels, stochastic block kernels and hub-like kernels, on ground sets of size 2 to 14 — for every pair with $m + n \leq 7$. The minimum of $a_{m+n} - a_m a_n$ over that sample was zero to machine precision, attained only at regular kernels. In spectral terms Conjecture A.2 is the correlation inequality $\int \lambda^{m+n} \geq \int \lambda^m \int \lambda^n$, which by the first remark above is false for general probability measures; a proof must therefore use more about W than its spectrum. We have no proof.

A.5 Consequence for the formalization

The Lean development splits cleanly at (10). Everything else is available, so the statement

$$p \geq \tfrac{1}{2} \text{ and } p^2\, t(P_3, W) \leq t(P_5, W) \implies p^4(2p-1) \leq t(R_3, W)$$

is unconditional Lean mathematics, and Atlas 102 becomes `verified` exactly when (10) is discharged — either by a direct analytic proof, or by formalizing the finite Blekherman–Raymond $(3,5)$ theorem together with the sampling transfer of Lemma 2.2. The second route is much the heavier: it needs a product probability space, a conditional Bernoulli construction, a second-moment estimate and convergence in probability, none of which the catalogue currently has.

B What the Lean formalization actually proves

Theorem 3.2 is formalized unconditionally, as `satisfiesLowerBound_102` in `lean/Taeyoung/Methods/OddWalk/Row`
Atlas 102 is `verified`. Two differences of substance.

Lemma 2.2 is discarded, and its gap is closed

Appendix A leaves (10) open and weighs two ways to obtain it, calling the second — the finite Blekherman–Raymond theorem plus the sampling transfer proved in Lemma 2.2 — much the heavier, since it needs a product probability space, a conditional Bernoulli construction, a second-moment estimate and convergence in probability.

None of that is formalized. The proof of Lemma 2.2 given above is not used at all: the $(3,5)$ inequality $a_3^5 \leq a_5^3$ is instead proved outright, by the entropy argument of `notes/blekherman_raymond.pdf` §2 run against relative entropy (see the appendix of that note for how). (10) then follows in two lines, since $a_3 \geq p^3$ gives $(p^2 a_3)^3 = p^6 a_3^3 \leq a_3^5 \leq a_5^3$.

The theorem is proved through Appendix A, not through Section 3

The auxiliary probability measure ν, the Jensen step and the function $f(a) = \max\{2a - p, 0\}$ of Theorem 3.2's proof do not occur. What is formalized is (8) followed by Lemma A.1; the full odd-walk inequality is used only through the weaker ratio (10).

References

[1] G. Blekherman and A. Raymond, *A new proof of the Erdős–Simonovits conjecture on walks*, Graphs and Combinatorics 39 (2023), Article 53; arXiv:2009.10845.